Exponentially Weighted Moving Average (EWMA)

1. Introduction

The **Exponentially Weighted Moving Average (EWMA)** is a concept used in many areas — statistics, finance, and **deep learning optimization**.

In Deep Learning, EWMA helps **smooth out fluctuations** and **capture trends** — especially in gradients or losses during training.

In simple words:

EWMA gives *more importance to recent values* and *less to older ones*.

That's why we call it "**exponentially weighted**."

2. What is EWMA?

Let's understand this with a simple example

Imagine you're tracking your **daily temperature** for 10 days.

Some days are hot, some are cold — so the data jumps up and down.

If you take a **simple average**, every day gets *equal weight* — even old data.

But we want recent temperatures to matter **more**.

So we use **Exponentially Weighted Average**:

- It smooths the curve.
- Recent points affect the average more strongly.
- Old points slowly "fade away" in importance.

This helps us see **the general trend** rather than noisy ups and downs.

3. Mathematical Formula

Let's denote:

- v_t: exponentially weighted average at time step t
- θ_t: the actual new data point (e.g., loss, gradient, etc.)
- β: smoothing factor (between 0 and 1)

Then the formula is:

$$v_t = \beta v_{t-1} + (1 - \beta)\theta_t$$

➢ Intuition Behind the Formula

- v_{t-1}: previous average
- θ_t: current value (new observation)
- β: controls how much weight we give to the past

- Higher (beta) → more weight on old data → smoother curve but slower to react
- Lower (beta) → more weight on recent data → more responsive but noisier

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➢ Example

Let's take:

- $\beta = 0.9$
- previous average $v_{t-1} = 10$
- new value $\theta_t = 20$

Then:

$$v_t = 0.9(10) + 0.1(20) = 9 + 2 = 11$$

So, instead of jumping straight to 20, the average moves *slightly* to 11.
It *smooths* the change.

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4. Mathematical Intuition

If you keep expanding the equation, you'll notice this:

$$v_t = (1 - \beta)(\theta_t + \beta\theta_{t-1} + \beta^2\theta_{t-2} + \beta^3\theta_{t-3} + \dots)$$

This means:

- The most recent value (theta_t) has the **highest weight** (multiplied by (1-\beta)).
- Older values have **exponentially smaller weights** (beta^2, beta^3, ...).

Hence, the name **Exponentially Weighted**.

➢ Bias Correction (Optional Concept)

In early steps (when (v_0 = 0)), the values are *biased* toward zero.
To fix that, we use **bias correction**:

$$v_t^{corrected} = \frac{v_t}{1 - \beta^t}$$

This is used in optimizers like **Adam** and **RMSProp**, so the moving averages start correctly even at the beginning of training.

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5. Why EWMA is Important in Deep Learning

EWMA is **not just for graphs**, it's a **core idea in optimization algorithms** like:

- **Momentum Optimizer**
- **RMSProp**
- **Adam**

These optimizers use EWMA to **track moving averages** of gradients and their squares, which helps:

- Smooth noisy gradients
- Stabilize weight updates
- Speed up convergence

In short:

- Momentum uses EWMA of **gradients** → remembers direction
- RMSProp uses EWMA of **squared gradients** → adapts learning rate
- Adam combines both EWMA types

So EWMA = *the secret ingredient* in making optimizers smarter.

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7. Summary

Concept	Description
Purpose	Smooths values by giving more weight to recent data
Formula	$v_t = \beta v_{t-1} + (1 - \beta)\theta_t$
β (beta)	Smoothing factor (closer to 1 = smoother curve)
Used In	Momentum, RMSProp, Adam
Benefit	Reduces noise, stabilizes learning, tracks trends

➢ Quick Analogy:

Imagine you're keeping track of your daily test scores.
Instead of taking a simple average of *all* past tests, you care more about how you're doing *recently*.
EWMA helps you focus on **recent performance**, while still remembering the past a little — that's exactly how optimizers "remember" recent gradients.
